

Analysis and Comparison of Node Rankings from PageRank and HITS

Git Hub Repo- https://github.com/Manyagoel14/IR_Assignment

Paper Title	In Top Page Rank	In Top Authority	In Top Hub
Dynamic Graph Representation Learning for Social Influence	Rank1	No	Rank1
DeepInf: Social Influence Prediction with Deep Learning	Rank2	No	Rank2
Graph Convolutional Networks for Social Recommendation	Rank3	No	Rank3
Neural Influence Modeling in Social Networks	Rank4	Rank3	Rank4
Representation Learning for Influence Diffusion	Rank5	Rank2	Rank5

Interpretation

- The PageRank and Hub rankings are **identical across the top three papers**, suggesting that the **most connected documents** in the network are also the ones **actively linking to other influential** documents.
- **Widely cited papers** also **engage** deeply with **other important** studies, forming a tightly interconnected core of research.
- The paper Dynamic Graph Representation Learning for Social Influence holds Rank 1 in both PageRank and Hub scores, making it the most central and influential paper in the dataset.
- In contrast, Neural Influence Modelling in Social Networks and Representation Learning for Influence Diffusion stand out because they **appear as both hubs and authorities**. This means they not only **cite several key** documents but are also frequently cited by others.

Conclusion

- PageRank identifies papers with the greatest overall influence.
- HITS (Hyperlink-Induced Topic Search), by distinguishing authorities from hubs, highlights how Neural Influence Modelling and Representation Learning for Influence Diffusion serve dual roles, both as sources of established knowledge (authorities) and as active connectors that integrate ideas from multiple influential papers (hubs).
- PageRank focuses on **influence and prominence**, while HITS captures the **flow and exchange of knowledge** within the network.